

SENTHAMIZH SELVAN S

Architectural BIM Modeler

senthamizh1031@gmail.com | +91 96775 85031

[Linkedin](#) | [Portfolio](#)

Passport Number : Y3467307

PROFESSIONAL SUMMARY

BIM Modeler with 2.5 years of professional experience delivering CAD-to-BIM and Scan-to-BIM projects in Autodesk Revit, with models developed from LOD 100 up to LOD 400 across commercial, residential, fit-out, heritage-renovation and landscape projects. Experienced in converting 2D CAD drawings into coordinated, data-rich architectural and structural BIM models. Proficient in producing GA plans, sections, elevations, ceiling and flooring plans, schedules and detailed construction drawings directly from the model. Demonstrated expertise in BIM Execution Plans (BEP), LOD/LOI management, QA/QC, clash detection in Navisworks, and Common Data Environment (CDE) workflows aligned with ISO 19650. Developed pyRevit and Python automation tools, including a point-cloud-to-Revit wall generation pipeline, to reduce repetitive modeling and documentation time. Currently delivering as-built Scan-to-BIM models from point cloud, Matterport and drone data. Seeking BIM-driven architectural teams where technical excellence and coordination are critical to project success.

KEY BIM SKILLS

- **Revit-Based Architectural Modelling & Documentation**, producing coordinated GA drawings, plans, sections, elevations, schedules directly from BIM models, while enforcing quality standards.
- **Development and Management of BIM Models** (existing-condition, proposed, and coordinated) in accordance with project-specific BIM Execution Plans (BEP), with QA/QC checks to maintain model accuracy and reliability.
- **Interdisciplinary BIM Coordination** with MEP teams, including clash detection, issue resolution, and proactive design coordination to prevent delays and cost overruns.
- **Scan-to-BIM Modeling**, converting .rcs point clouds, Matterport panoramas and DroneDeploy project data into accurate as-built Revit models at LOD 300 for existing retail buildings.
- **Preparation and Management of Project Submissions** such as RFIs, design clarifications, and coordinated submittals to ensure smooth communication across all stakeholders.
- **Project Clash Coordination** using Navisworks & QA/QC for the projects, including new-build and renovation works, ensuring design integrity and compliance at all stages.
- **BIM Workflow Automation** using PyRevit & Python Revit API, developing custom plugins to automate repetitive modeling and documentation tasks, enhancing model management efficiency and reducing coordination time.

PROFESSIONAL EXPERIENCE

Junior Executive Engineer / BIM Modeler

Trichy, Tamil Nadu, India

Conserve Solutions.

Sep 2026 - Present

- Delivered **Scan-to-BIM** projects for the **Cumberland Farms** convenience store chain, modeling existing buildings in Autodesk Revit at **LOD 300** from the supplied survey inputs.
- Developed **as-built BIM models from .rcs point cloud files, Matterport panoramas and DroneDeploy project data**, cross-referencing all three sources to resolve geometry, openings, finishes and roof conditions.
- Modeled the **architectural, structural, mechanical, electrical and fire disciplines** for each building as a single coordinated as-built model.
- Produced documentation directly from the model, including **general arrangement plans, flooring plans, ceiling plans, elevations, sections and interior room plans**.
- Developed a **Python and pyRevit point-cloud-to-Revit wall generation pipeline** to automate part of the manual tracing workflow (see Automation & Custom Development).

- Developed and coordinated architectural and structural BIM models using Autodesk Revit in compliance with **BIM Execution Plans(BEP)**, project BIM standards, ISO 19650 guidelines, and client-specific requirements for commercial, residential, and fit-out projects.
- Delivered accurate BIM models across **LOD 100-400** ensuring consistency of geometry, information richness, and compliance with **LOD/LOI** requirements at each design and construction stage.
- Prepared comprehensive construction documentation including general **arrangement plans, sections, elevations, reflected ceiling plans, flooring plans, schedules, interior design drawings, millwork drawings and detailed technical drawings** directly from coordinated BIM models
- Coordinated architectural models with **Structural, MEP consultants**, resolving spatial and system conflicts through structured model reviews.

AUTOMATION & CUSTOM DEVELOPMENT

- **Developed Scan-to-BIM Wall Generation Pipeline (Python + pyRevit)**, a six-stage workflow using only free tools beyond licensed Recap and Revit: loads and cleans .las/.e57 point clouds, finds floor and ceiling levels from a Z-height histogram, slices the cloud at multiple heights to isolate clean wall geometry, builds a binary plan image, detects wall centerlines and creates real parametric Revit Wall elements aligned to the linked point cloud.
- **Developed Part Plan Generator** using filled region boundaries as automated crop regions with built-in template selector to rapidly create general arrangement, flooring, furniture, and specialty partial plans with custom naming conventions.
- **Created Annotation Transfer Tool** to copy and migrate room tags, spot elevations, flooring tags, and door tags across views with different scales, enabling efficient annotation replication for part plan documentation.
- **Built Room View Generator** to automatically create general arrangement, flooring, and ceiling plan views for individual rooms using room boundary geometry with customizable templates for room-specific sheet sets.
- **Designed Excel Parameter Updater** to batch synchronize element classification data including Uniclass codes, clash groups, and coordination priority values from spreadsheets to project models.

PROJECTS SUMMARY

- Design and Build for the Renovation of the Heritage Building of the Qatar Media Corporation, Qatar
FM Building ,AM Building And QTV Building - **LOD 100 - 300**
- Cumberland Farms Convenience Store Chain, Scan-to-BIM - **LOD 300 / as-built Models**
- Commonwealth Bank of Australia, (CBA) - **LOD 400**
- Project Realm - Qiddiya Project (Landscape) - **LOD 400**
- The Red Sea, Saudi Arabia - **LOD 400**
- Aldar Manarat, Dubai - **LOD 350**

SOFTWARE & SKILLS

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|------------------|-----------------|----------------------|-----------------|
| • Autodesk Revit | • BIM 360 / ACC | • Autodesk Recap Pro | • pyRevit |
| • AutoCAD | • Matterport | • Enscape | • Python Basics |
| • Navisworks | • Dronedeploy | • Microsoft Office | • Revit API |

EDUCATION

Bachelor of Engineering (Civil Engineering) – With Honors

CGPA: 8.46 / 10

Hindusthan College of Engineering and Technology, Coimbatore, India,

2020 - 2024

LANGUAGES

- English (Fluent - Professional Working Proficiency)
- Tamil (Native)